

enter the sleep mode and by releasing resources. This occurs in the event of turning off the device screen. This listener ensures the proper functioning of the device when resource-heavy applications with 3D graphics are being executed.

The `Osp::Base::Runtime::ITimerEventListener` interface is declared for listening to timer events. The spinning event of the 3D cube in this application occurs every time the timer expires.

```
• class GlesCube:
•     public Osp::App::Application,
•     public Osp::System::IScreenEventListener,
•     public Osp::Ui::IKeyEventListener,
•     public Osp::Base::Runtime::ITimerEventListener
• {
```

c. Finally, a set of OpenGL® and EGL™-specific variables is declared.

```
•     private:
•         Osp::Graphics::Opendgl::EGLDisplay __eglDisplay;
•         Osp::Graphics::Opendgl::EGLSurface __eglSurface;
•         Osp::Graphics::Opendgl::EGLConfig __eglConfig;
•         Osp::Graphics::Opendgl::EGLContext __eglContext;
•         Osp::Graphics::Opendgl::GLuint __programObject;
•         Osp::Graphics::Opendgl::GLint __idxPosition;
•         Osp::Graphics::Opendgl::GLint __idxColor;
•         Osp::Graphics::Opendgl::GLint __idxMVP;
•         Osp::Graphics::Opendgl::GLfloat __angle;
•         Osp::Base::Runtime::Timer* __pTimer;
•         Matrix __matMVP;
•
•         Osp::Ui::Controls::Form* __pForm;
• };
```

2. GlesCube.cpp Source File

a. OpenGL® ES 2.0 or later uses the shader model. A shader is a set of GPU (Graphics Processing Unit) instructions that is executed on the graphics hardware to enhance the final result on the screen. A fragment shader is used to create per-pixel effects on the cube, and a vertex shader is used to create a special effect each time a vertex of the cube is processed.

The fragment.h and vertex.h files are included in the GlesCube.cpp file to add defined shaders for the 3D cube. Both files contain an array of shader information.